

Muhendislik Matematigi - Vize Sinavi Cozumleri

Cozum 1:

Kismi turevler: $f_x = 3x^2 - 3 = 0 \Rightarrow x^2 = 1 \Rightarrow x = 1$ veya $x = -1$

$f_y = 2y - 6 = 0 \Rightarrow y = 3$

Kritik noktalar: $(1, 3)$ ve $(-1, 3)$

Ikinci merteye turevler: $f_{xx} = 6x$, $f_{yy} = 2$, $f_{xy} = 0$

$D = (f_{xx})(f_{yy}) - (f_{xy})^2 = (6x)(2) - 0 = 12x$

$(1, 3)$ icin: $D = 12 > 0$ ve $f_{xx} = 6 > 0 \Rightarrow (1, 3)$ yerel minimum noktasidir.

$(-1, 3)$ icin: $D = -12 < 0 \Rightarrow (-1, 3)$ eyer noktasidir.

Cozum 2:

$\det(A - \lambda I) = (4 - \lambda)(1 - \lambda) - (-2)(1) = \lambda^2 - 5\lambda + 4 + 2 = 0$

Karakteristik denklem: $\lambda^2 - 5\lambda + 6 = 0 \Rightarrow (\lambda - 2)(\lambda - 3) = 0$

Ozdegerler: $\lambda_1 = 2$, $\lambda_2 = 3$

$\lambda_1 = 2$ icin ozvektor: $(A - 2I)v = 0 \Rightarrow \begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow x - y = 0 \Rightarrow v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}^T$

$\lambda_2 = 3$ icin ozvektor: $(A - 3I)v = 0 \Rightarrow \begin{bmatrix} 1 & -2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow x - 2y = 0 \Rightarrow v_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}^T$

Cozum 3:

Homojen cozum (y_h): $r^2 + 4 = 0 \Rightarrow r = \pm 2i \Rightarrow y_h = C_1 \cos(2x) + C_2 \sin(2x)$

Ozel cozum (y_p): $y_p = Ax^2 + Bx + C$ diyelim. $y_p' = 2Ax + B$, $y_p'' = 2A$

Yerine koyelim: $2A + 4(Ax^2 + Bx + C) = 8x^2 \Rightarrow 4Ax^2 + 4Bx + (2A + 4C) = 8x^2$

Katsayilari esitleyelim: $4A = 8 \Rightarrow A = 2$; $4B = 0 \Rightarrow B = 0$; $2A + 4C = 0 \Rightarrow 4 + 4C = 0 \Rightarrow C = -1$

Genel cozum: $y = y_h + y_p = C_1 \cos(2x) + C_2 \sin(2x) + 2x^2 - 1$

Cozum 4:

lc integral: $\int_0^1 xy \, dx = \left[\frac{x^2}{2} y \right]_0^1 = \frac{y^2}{2} \Big|_0^1 = \frac{y^3}{2} \Big|_0^1 = \frac{1^3}{2} - 0 = \frac{1}{2}$

Dis integral: $\int_0^2 \left(\frac{y^3}{2} \right) dy = \left[\frac{y^4}{8} \right]_0^2 = \frac{16}{8} - 0 = 2$